Latency

p99, latency budgets and the numbers to know

Watch the 3-min video

In one line: latency is the time for one request, made of network, queueing, processing and I/O; measure it as percentiles, know the key numbers, and split the target into a budget per component.

1 What latency is made of

Latency is the time from sending a request to receiving the full response. It's never one thing; it's the sum of four parts, and each part has different fixes.



latency = network + queueing + processing + I/O

Figure 1. The four parts of latency. Queueing is the sneaky one: it grows fast near full utilization, even when your code doesn't get slower.

2 Measure percentiles, not averages

p50 is the typical request; **p95** and **p99** are the slow tail. If one page calls 10 services, the chance that at least one call is at or beyond its p99 is $1 - 0.99^{10} \approx$ **10%**. With 50 calls it's ~40%. The tail is what users actually feel.

3 Numbers every engineer should know

Operation	Typical time	Relative to memory
Read from main memory	~100 ns	1×
Random read from an SSD	~100 µs	~1,000×
Round trip inside one data centre	~0.5 ms	~5,000×
Read 1 MB sequentially from an SSD	~1 ms	~10,000×
Round trip across regions (e.g. India ↔ Europe)	100–150 ms	~1,000,000×

Rough orders of magnitude are enough: they tell you instantly that a cross-region call inside a loop is a bad idea, and that caching in memory beats any disk.

4 Latency budgets

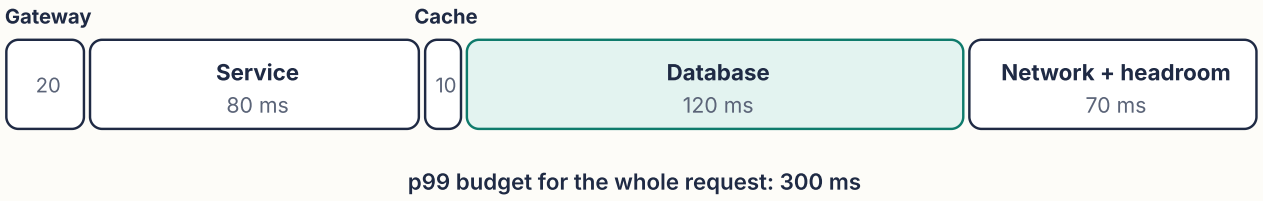


Figure 2. A 300 ms p99 target split into a budget per component. Every team knows its number.

If a part goes over budget, you know where to look, and each fix maps to one part of latency:

Too much ...	Fix
Network	Move closer to users (CDN, regions), cut round trips, batch calls
Queueing	Add capacity, lower utilization, shed low-priority load
Processing	Profile hot code, do less work per request, move work to async jobs
I/O / database	Add an index, cache hot reads, avoid N+1 queries

COMMON MISTAKES

- Reporting average latency.
- Calling services one after another when they could run in parallel.
- Ignoring cross-region round trips in the design.
- Measuring latency without stating the load it was measured at.

5 Key terms

Latency: time for one request

Tail latency: the slowest few percent

Latency budget: max time per component

p99: 99% of requests are faster

Queueing delay: time waiting for a free worker

RTT: round-trip time

6 Interview questions

Q1. Why use p99 instead of the average? BASIC

Averages hide the slow tail, and pages make many calls, so most users hit some call's p99 regularly. Percentiles show what users actually feel.

Q2. How does a cross-region round trip compare with a memory read? BASIC

Roughly 100 ms versus 100 ns: about a million times slower.

Q3. What are the parts of latency? BASIC

Network time, queueing (waiting for a worker), processing (your code) and I/O (disks, databases). Each has different fixes.

Q4. Your API's p99 jumped from 120 ms to 900 ms but code didn't change. Where do you look? SCENARIO

Queueing and dependencies: higher utilization, a saturated pool or a slow database. Check traffic, CPU, pool wait time and database latency, using traces to see which span grew.

Q5. Split a 300 ms p99 budget for a checkout request. SCENARIO

Gateway ~20 ms, service logic ~80, cache ~10, database ~120, and ~70 for network and headroom. Then each team owns its number.

Q6. A page calls 5 services one after another, each 60 ms. How do you speed it up? SCENARIO

Call independent services in parallel (≈ 60 ms instead of 300), cache results that change rarely, or have one aggregator call them inside the data centre.

Q7. With 10 calls per page, why do ~10% of pages hit a p99? TRICKY

Each call has a 1% chance of being in its slowest 1%. The chance at least one of 10 is: $1 - 0.99^{10} \approx 9.6\%$.

Q8. Can adding a cache make p99 worse? TRICKY

Yes, if misses add a network hop before the database, or the cache is overloaded or cold after a restart. Measure hit rate and miss latency, and warm caches after deploys.

Next: SD-A-009 · Throughput & Little's Law backendroadmap.com